

5 (long) Figure 5 shows the circuit of an amplifier.

(a) Calculate the values of R_1 and R_2 required to set the transistor operating point, such that $V_{GS} = -2$ V, $V_{DS} = 8$ V and $I_D = 0.2$ mA. It can be assumed that the transistor is ideal and that there is no gate current. [8]

(b) Draw the small-signal model for the circuit, assuming that the impedances of the capacitors are negligible at small signal frequencies. [5]

(c) Derive approximate expressions for the small-signal gain v_o/v_i and the output impedance of the circuit, assuming that the output circuit draws no current. Hence evaluate these expressions for the values calculated in (a), and taking $g_m = 1$ mS, and $r_d = 200$ k Ω . [8]

(d) What value of C_1 is required if the circuit is to have a -3 dB point of 15 Hz, with a 10 k Ω load connected to the output, and assuming that the reactance of C_2 is small compared to R_2 at this frequency? [5]

(e) Under the conditions of part (d), find the maximum and minimum voltages at the drain of the transistor and across the 10 k Ω load resistor if the input voltage is a sinusoid of peak-peak amplitude 100 mV at a frequency of 1 kHz. [4]

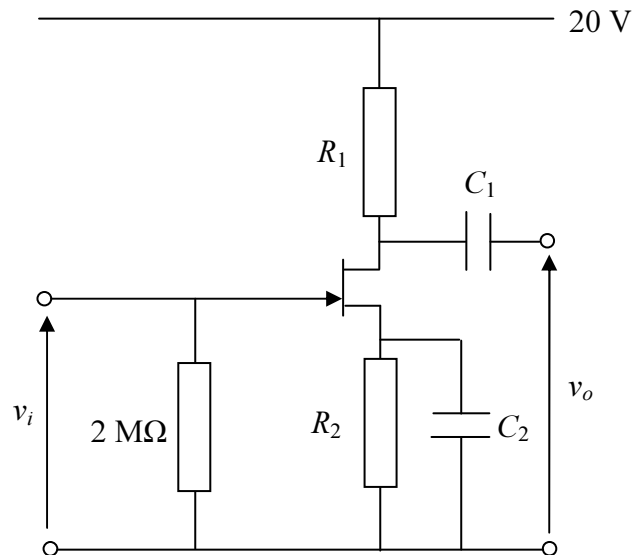


Fig. 5.